SANGHYEON NAM

Ph.D. Student | Controls, Robotics, and Autonomous Systems

snam@tamu.edu Google Scholar LinkedIn

Education

Texas A&M University, College Station, TX — Aug. 2025–Present
Ph.D. in Mechanical Engineering – Controls, Robotics, and Autonomous Systems

Kyungpook National University, Daegu, South Korea — Mar. 2017–Aug. 2024
B.S. and M.S. in Mechanical Engineering

Honors & Awards

- **James J. Cain '51 Graduate Student Award**, J. Mike Walker '66 Department of Mechanical Engineering, Texas A&M University, 2026; selected as an outstanding early-stage graduate student.
- Finalist, Fuel Cell Hybrid Electric Vehicle Control Competition, IFAC World Congress, 2023.
- Scholarship for Academic Excellence, KNU, 2017, 2022, and 2024; Academic Support Scholarship, 2022–2023.

Ph.D. Research & Innovation – Texas A&M University

Graduate Research Assistant – Autonomous Robotics, Control, and Learning — Aug. 2025–Present

ShaRE Project – Autonomous Agricultural Robotics

- Developed a camera-free and RTK/GNSS-free under-canopy navigation system using low-cost 2D LiDAR–IMU sensing, reliability-aware perception, predictive control, and a complete ROS2 perception-to-control autonomy stack for challenging crop-row environments.
- Developed a 2D LiDAR-based Adaptive Corridor Tightening MPC framework that converts perception reliability into dynamically adjusted navigation safety constraints.
- Demonstrated the system through autonomous traversals totaling 26.279 km in real cotton and corn fields, achieving zero navigation failures and zero operator interventions with repeated operation up to 0.8 m/s.

Learning-Enabled Control for Robotic Manipulators

- Developed a learning-enabled robust predictive-control framework for payload-sensitive robotic manipulators, achieving successful simulation validation under varying payload and obstacle conditions.

Prior Research Experience

Research Engineer – Korea Institute of Industrial Technology (KITECH) — Nov. 2024–Jul. 2025

- Developed safety-aware robust MPC, real-time obstacle avoidance, and collision-aware control for robotic manipulators in dynamic manufacturing environments, resulting in a first-author IEEE CASE 2025 publication.

Graduate Researcher – Kyungpook National University — Mar. 2023–Aug. 2024

- Developed hierarchical nonlinear MPC for electric-vehicle thermal management and robust/LMI-based autonomous-vehicle steering, with simulation and real-time validation, resulting in multiple journal and international conference publications.

Patents and Invention Disclosures

1. **S. Nam** and Kiju Lee, "Reliability-Aware LiDAR–IMU Under-Canopy Navigation and Adaptive Safety-Constrained Predictive Control Software for Agricultural Ground Vehicles", *Invention Disclosure*, filed September 2026.

Publications

In Preparation

1. **S. Nam**, Abdulaziz Farhan Alharbi, Muthu Bagavathiannan, and Kiju Lee, "Learning When to Trust Geometry: Reliability-Aware Stochastic MPC for Low-Cost Under-Canopy Navigation," manuscript in preparation for submission to ICRA 2027.
2. **S. Nam**, Abdulaziz Farhan Alharbi, and Kiju Lee, "Learning-Enabled Robust Predictive Control for Payload-Sensitive Robotic Manipulators," manuscript in preparation, 2026.

Refereed Journal Articles

3. **S. Nam**, H. Lee, Y. Kim, K. H. Kwak, and K. Han, "Hierarchical Climate Control Framework for Electric Vehicles Considering Door-Opening Event," *IEEE Transactions on Intelligent Vehicles*, 2025.
4. N. N. Nam, **S. Nam**, H. D. Nguyen, S. Kim, and K. Han, "Offline Robust Model Predictive Control for Autonomous Vehicle Steering Systems Using LMI-Based Optimization," *International Journal of Robust and Nonlinear Control*, 2026.

Refereed Full-Length Conference Papers

5. **S. Nam**, D. Kim, S.-H. Choi, C.-H. Kim, H. Kwon, H. Kawamoto, and S. Lee, "Safety-Aware Robust Model Predictive Control for Robotic Arms in Dynamic Environments," *IEEE International Conference on Automation Science and Engineering (CASE)*, 2025.

Teaching & Leadership

- Graduate Teaching Assistant – MEEN 364: Dynamic Systems and Control, Texas A&M University, Fall 2025; taught lecture sessions, led office hours, prepared solution materials, and supported course assessment.
- Mentored undergraduate researchers in vehicle-control projects and conducted hands-on control-system training using the Quanser QUBE Servo 2.

Technical Expertise

Robotics: ROS2, field robotics, autonomous navigation, 2D LiDAR, IMU, robotic manipulators
Control: MPC, SMPC, NMPC, robust MPC, learning-enabled control, constrained optimization
Programming/Testing: Python, C++, MATLAB/Simulink, Linux, CarSim, dSPACE, MicroAutoBox, Speedgoat, HIL